

class08

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class. You’ll extend what you’ve learned by combining PCA as a preprocessing step to clustering using data that consist of measurements of cell nuclei of human breast masses. This expands on our UK food and RNA-Seq analysis from last day.

Today we will analyze a data-set from fine needle aspiration (FNA) of breast mass

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
#wisc.df

head(wisc.df, 4) #examine input data to ensure column names are set correctly
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	
842302	M	17.99	10.38	122.80	1001.0	
842517	M	20.57	17.77	132.90	1326.0	
84300903	M	19.69	21.25	130.00	1203.0	
84348301	M	11.42	20.38	77.58	386.1	
		smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302		0.11840	0.27760	0.3001	0.14710	
842517		0.08474	0.07864	0.0869	0.07017	
84300903		0.10960	0.15990	0.1974	0.12790	
84348301		0.14250	0.28390	0.2414	0.10520	
		symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302		0.2419	0.07871	1.0950	0.9053	8.589
842517		0.1812	0.05667	0.5435	0.7339	3.398
84300903		0.2069	0.05999	0.7456	0.7869	4.585
84348301		0.2597	0.09744	0.4956	1.1560	3.445
		area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302		153.40	0.006399	0.04904	0.05373	0.01587
842517		74.08	0.005225	0.01308	0.01860	0.01340
84300903		94.03	0.006150	0.04006	0.03832	0.02058
84348301		27.23	0.009110	0.07458	0.05661	0.01867
		symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302		0.03003	0.006193	25.38	17.33	
842517		0.01389	0.003532	24.99	23.41	
84300903		0.02250	0.004571	23.57	25.53	
84348301		0.05963	0.009208	14.91	26.50	
		perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302		184.60	2019.0	0.1622	0.6656	
842517		158.80	1956.0	0.1238	0.1866	
84300903		152.50	1709.0	0.1444	0.4245	
84348301		98.87	567.7	0.2098	0.8663	
		concavity_worst	concave.points_worst	symmetry_worst		
842302		0.7119	0.2654	0.4601		
842517		0.2416	0.1860	0.2750		
84300903		0.4504	0.2430	0.3613		
84348301		0.6869	0.2575	0.6638		
		fractal_dimension_worst				
842302		0.11890				
842517		0.08902				
84300903		0.08758				
84348301		0.17300				

Make sure we remove or exclude the **diagnosis** column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]

# Create diagnosis vector for later
diagnosis <- as.factor(wisc.df$diagnosis)
```

2. Exploratory data analysis

>Q1. How many observations are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

569 observations in this dataset.

>Q2. How many of the observations have a malignant diagnosis?

```
table((wisc.df$diagnosis == "M"))
```

```
FALSE  TRUE
   357   212
```

212 observations have a malignant diagnosis.

>Q3. How many variables/features in the data are suffixed with `_mean`?

```
length( grep("_mean", colnames(wisc.data), value = T))
```

```
[1] 10
```

10 variables in the data are suffixed with `_mean`.

3. Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`

```
# Check column means and standard deviations  
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se

1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
# Perform PCA on wisc.data
wisc.pr <- prcomp( wisc.data, scale = TRUE )

summary(wisc.pr) # summary of results
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Performing PCA

>Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

.4427, 44.27% of the original variance is captured by the PC1.

>Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 principal components are required to describe at least 70% of the original variance in the data.

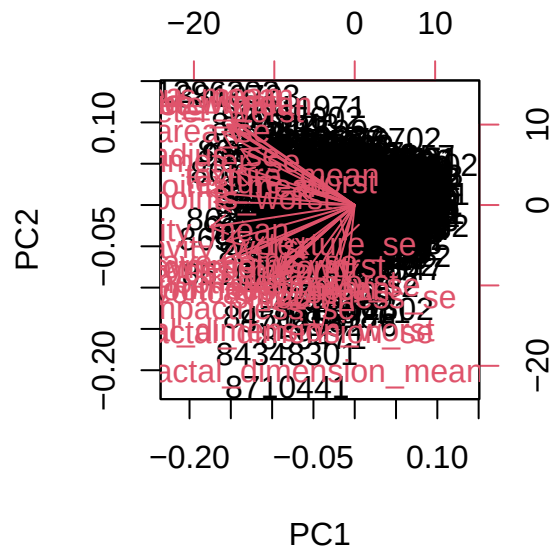
>Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 principal components are required to described at least 90% of the original variance in the data.

Interpreting PCA results

Biplot

```
biplot(wisc.pr)
```



> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

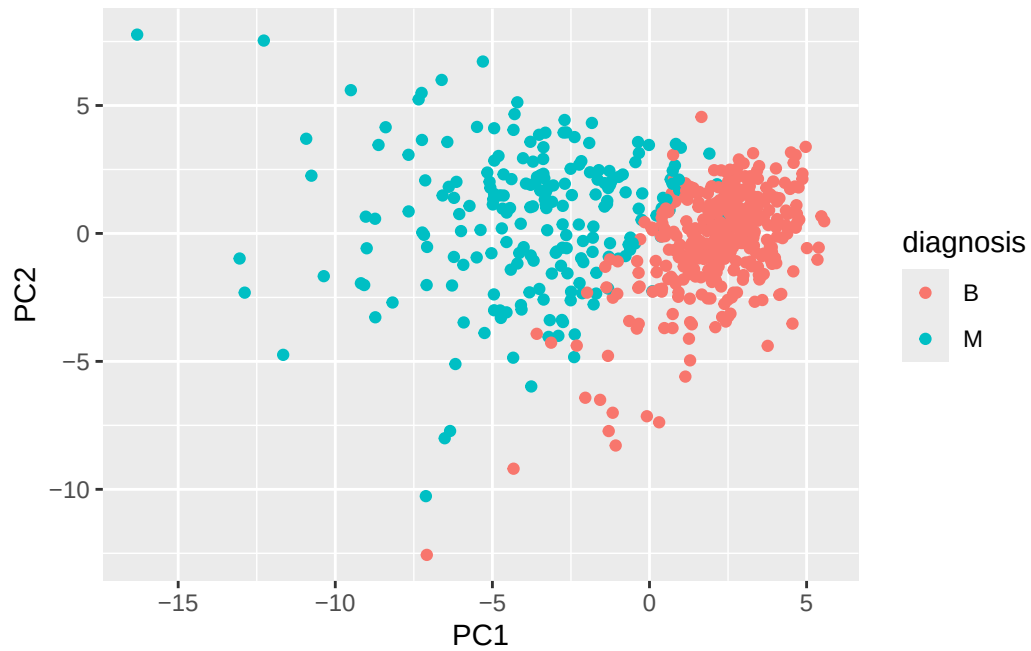
Nothing really stands out. It is difficult to understand because there are many values on a small plot, making it hard to see any values and interpret them.

Let's see the "PC score plot"

```
library(ggplot2)
```

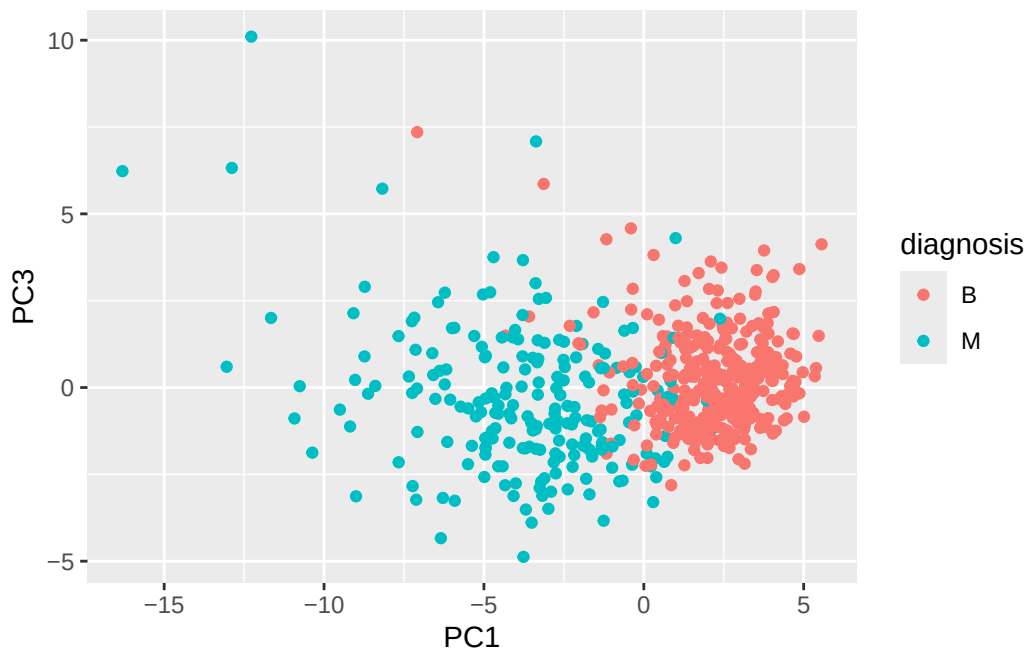
Warning: package 'ggplot2' was built under R version 4.5.2

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC2, col=diagnosis) +  
  geom_point()
```

> **Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?**

```
# Repeat for components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



I notice that PC1 captures the separation between malignant(blue) and benign(red) samples well, but PC3 allows for overlapping and less separation on the PC3 axis, making it harder to distinguish. PC2 draws the sample a little more apart, allowing for more separation and variance.

Variance Explained

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

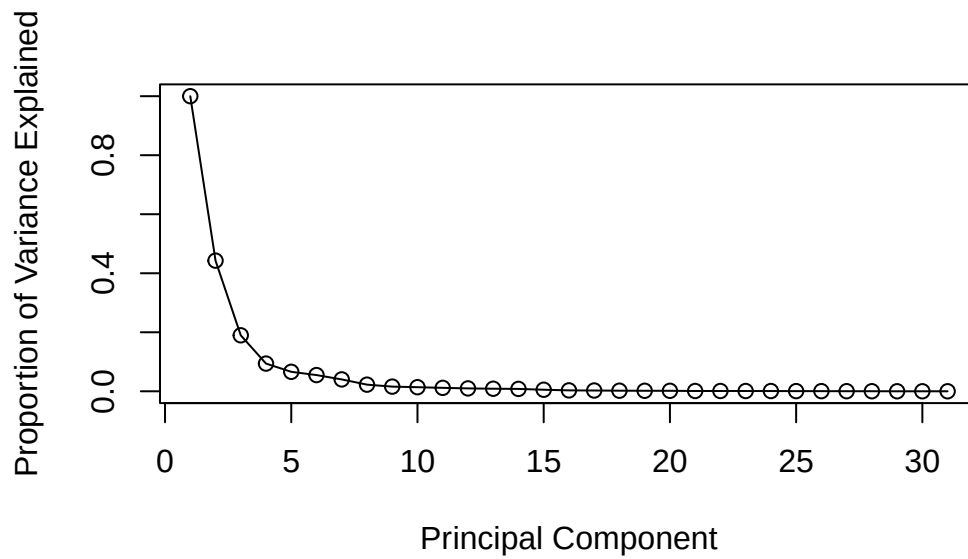
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Calculate the variance explained by each principal component by dividing by the total variance explained of all principal components. Assign this to a variable called pve and create a plot of variance explained for each principal component.

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

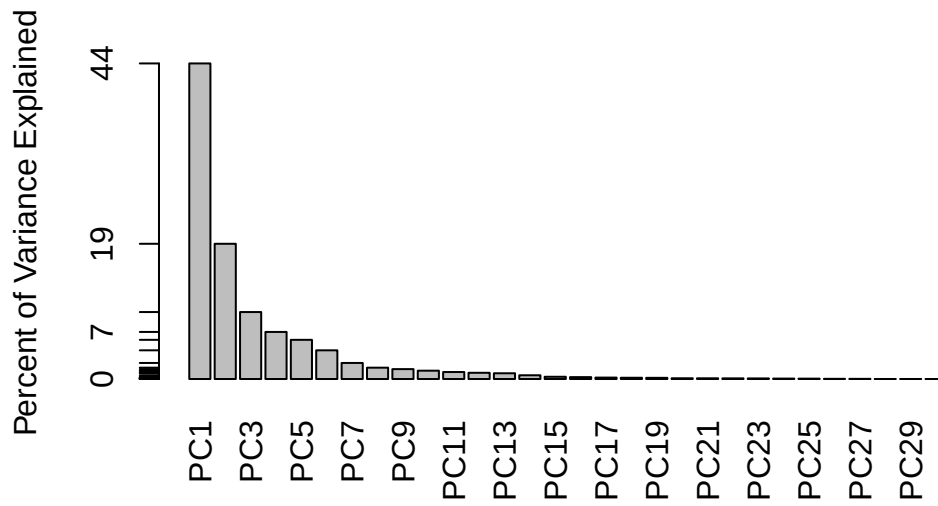
# Plot variance explained for each principal component
```

```
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Alternative scree plot

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

> **Q9.** For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se

0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

The component of the loading vector for the feature `concave.points_mean` is -0.2609. No, there are not any features with a larger absolute contribution than this one for PC1.

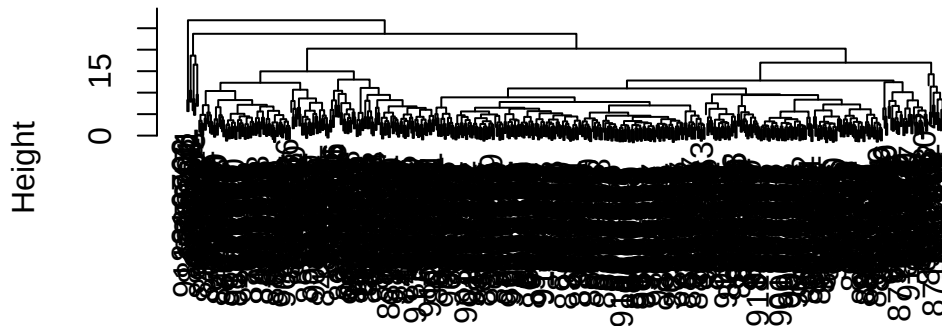
4. Hierarchical Clustering

```
data.scaled <- scale(wisc.data) #scale wisc.data data

data.dist <- dist(data.scaled) #calculate (Euclidean) distances between all pairs of observations

#create hierarchical clustering model using complete linkage
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```

Cluster Dendrogram

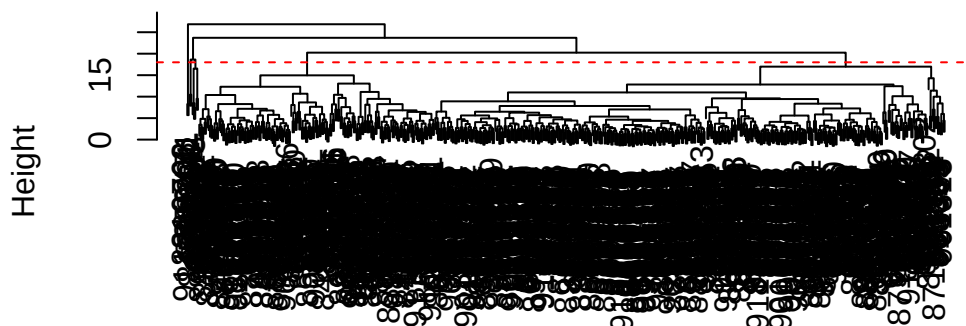


```
dist(scale(wisc.data))  
hclust (*, "complete")
```

##Results of hierarchical clustering ### > Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)  
abline(h=18, col="red", lty=2)
```

Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "complete")
```

At height 19, the clustering model has 4 clusters.

Using Different Methods

Selecting number of clusters

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)  
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

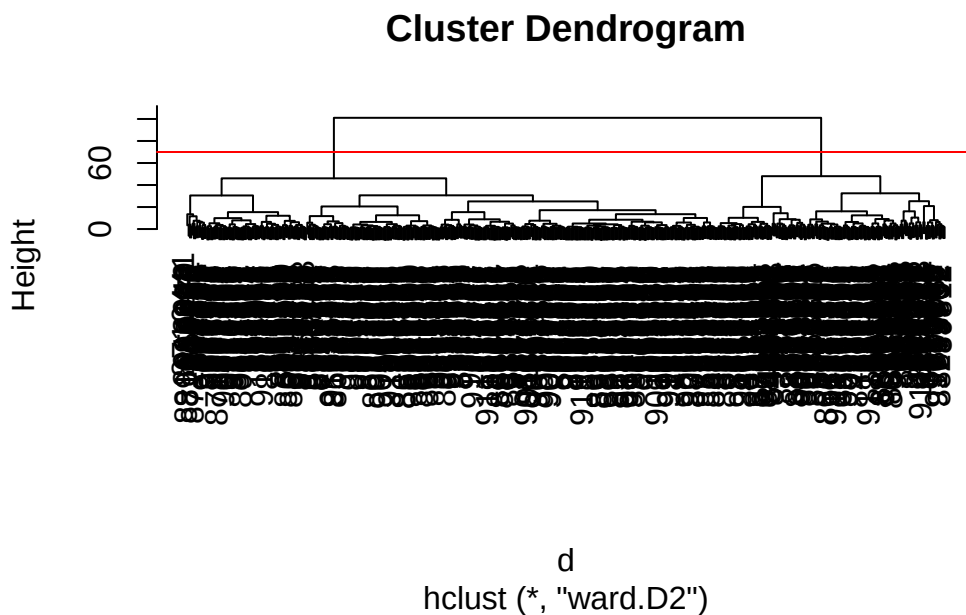
**> Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.**

My favorite method is the “ward.D2” because it is the most organized looking to me. It makes compact and clean clusters, and looks neater compared to the other methods, which can be helpful when I may need to analyze it.

5. Combining Methods

Clustering on PCA results

```
d <- dist(wisc.pr$x[,1:4])  
wisc.pr.hclust <- hclust(d, method = "ward.D2") #run hierarchical clustering on d  
plot(wisc.pr.hclust)  
abline(h=70, col="red")
```



```
grps <- cutree(wisc.pr.hclust, h=70)  
table(grps)
```

```
grps  
  1  2  
171 398
```

How does this clustering `grps` correspond to the expert diagnosis?

```
table(diagnosis)
```

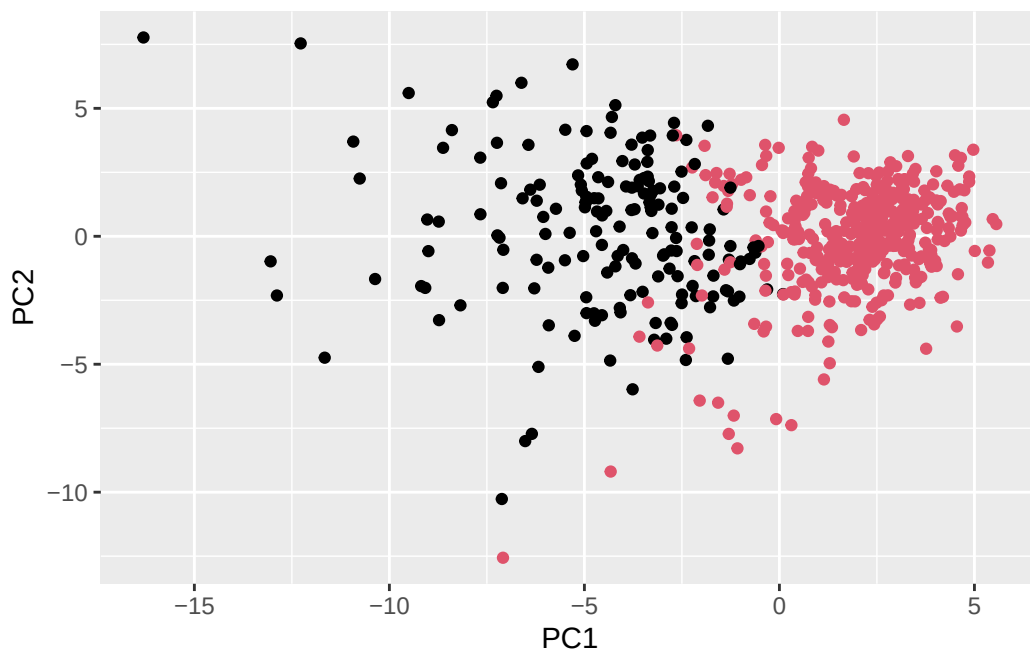


```
diagnosis
  B   M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1   2
  B      6 351
  M     165  47
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
pc.dist <- dist(wisc.pr$x[, 1:7])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")

## Cut this hierarchical clustering model into 2 clusters and assign the results to wisc.pr.
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

Clustering on PCA results

> **Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?**

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	28	188
2	329	24

The newly created hclust model separates malignant and benign samples pretty well. Cluster 1 contains mostly malignant cases (188 M vs 28 B), while Cluster 2 contains mostly benign cases (329 B vs 24 M).

> **Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.**

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

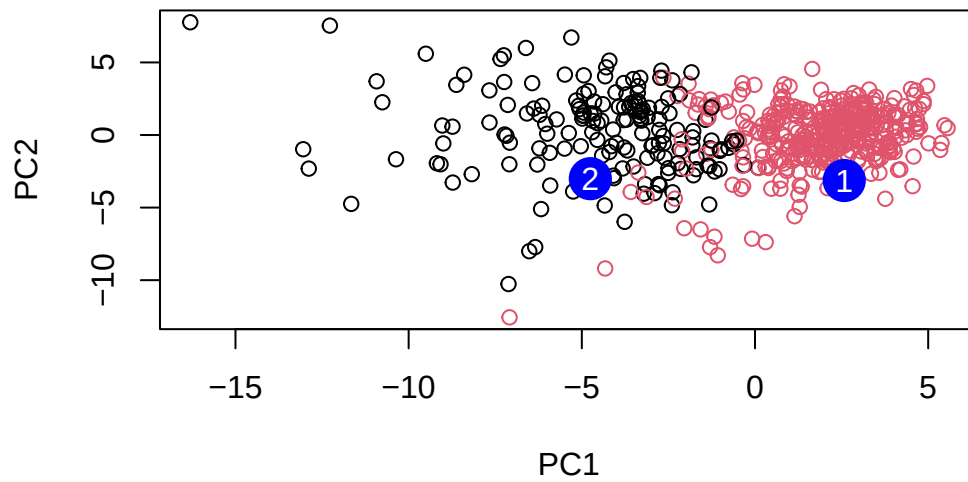
The hierarchical clustering model without first doing PCA separates the diagnoses well, where one cluster is mostly malignant (cluster 1) and another mostly benign (cluster 3), but it also produces many small extra clusters (cluster 2 and 4). In contrast, the PCA-based clustering gives two larger, cleaner clusters that better correspond to the two diagnosis groups, in which “with doing PCA first” improves the understanding and comprehension of the clustering and separation overall.

7. Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



> Q16. Which of these new patients should we prioritize for follow up based on your results?

Based on my results, we should be prioritizing Patient 2 because it falls closer to the group of malignant samples (black group) and Patient 1 falls within the benign cluster samples (red group).